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Contents

Abstract **vii**
Sommario ix

1 Introduction **1**
1.1 Motivation 1
1.2 Energy Harvesting Systems 2
1.3 Performance metrics 3
1.4 Main Transduction Technologies 4
1.5 The Variable Reluctance Principle 5
1.6 State of the Art of VREH 6
1.7 Open Challenges 7
1.8 Harvester Specifications and Research Objective 9

2 Theoretical Background **11**
2.1 Electrical Model of the Variable-Reluctance Harvester 11
2.1.a Faraday's law 11
2.1.b Variable Reluctance Principle 12
Pickup-unit topology and flux-improvement strategies . . 16

3 Rectifiers **19**

Bibliography **21**

Acknowledgments **25**

Abstract

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1 Introduction

1.1 Motivation

The number of wireless sensor networks (WSNs) and Internet of Things (IoT) devices has grown rapidly in recent years, raising a fundamental question: how to power all these devices without replacing batteries or requiring constant maintenance. Recent advances in low-power electronics have reduced the energy consumption of sensor nodes to the range of tens to hundreds of microwatts [10]. At such low power levels, it becomes possible to power these devices by extracting energy from the surrounding environment, removing the need for batteries and allowing the network to operate indefinitely.

Batteries are the most common power source for WSN nodes, but they have well-known limitations. In many applications, such as structural health monitoring, industrial machinery surveillance, or medical implants, replacing or recharging depleted batteries is difficult or simply not possible [5]. Using larger batteries to extend the device lifetime increases the size of the system, which is often not acceptable. In addition, the environmental cost of battery and the consequent maintenance of large-scale networks are serious obstacles to the widespread use of WSNs [8].

Energy harvesting offers a practical solution to these problems. Instead of storing a fixed amount of energy, harvesting systems collect energy from sources available in the environment such as sunlight, heat gradients, mechanical vibrations, or radio frequency signals and convert it into electrical power to supply an electronic system.

Among all available energy sources, mechanical vibrations have received the most attention in the research community, accounting for more than half of publications in this field [8]. Piezoelectric, triboelectric, and electromagnetic transducers are the most widely used mechanisms, each with different trade-offs in output voltage, power density, and complexity [10, 5]. Figure 1.1 shows the distribution of recent publications about transducer types, highlighting the dominance of mechanical energy harvesting approaches.

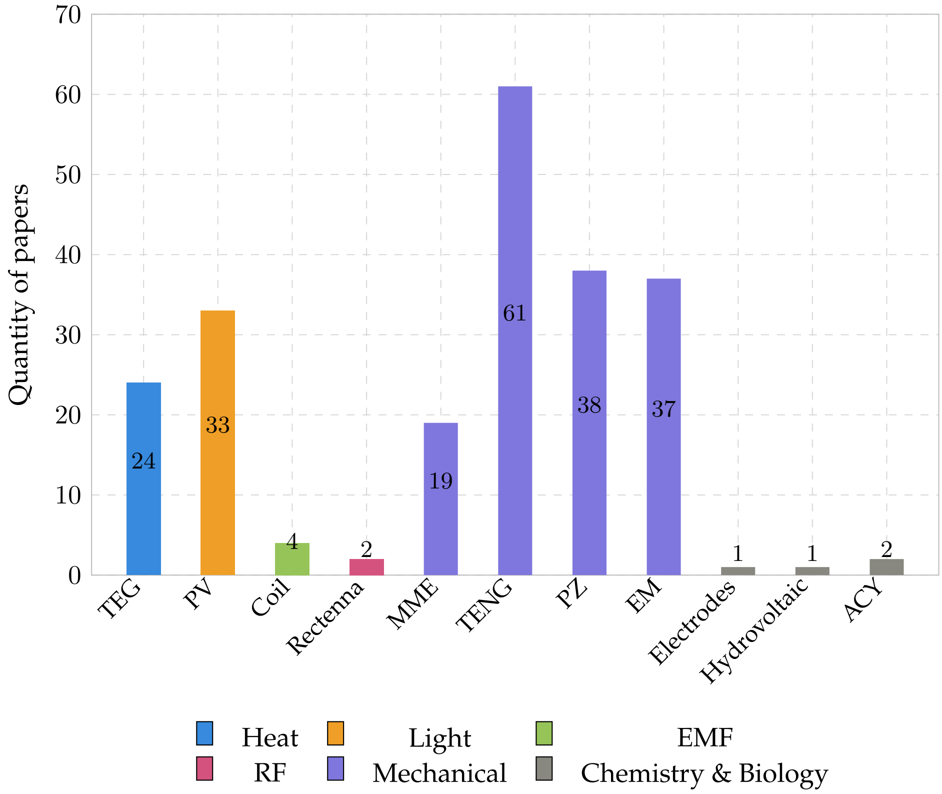

Figure 1.1: Distribution of papers based on transducer type [8].

Energy harvesting systems are generally divided into two categories [5]: systems that use the harvested energy immediately (*Harvest-Use*), and systems that store it first in a battery or supercapacitor before supplying it to the node (*Harvest-Store-Use*). The second approach is far more common, since most energy sources are not continuously available and some form of storage is needed to guarantee uninterrupted operation.

1.2 Energy Harvesting Systems

An energy harvesting system is made of three components that work in sequence, as illustrated in Figure 1.2.

- *Transducer*: converts the ambient energy into an electrical signal.
- *Power conditioning circuit*: perform AC-DC conversion, impedance matching and voltage regulation. A typical implementation combines a rectifier circuit with a power management integrated circuit (PMIC) [14].
- *energy storage element*: is used to keep the device operating for a certain amount of time even if the source is week or no more present. Such element can be a capacitor/super capacitor or a rechargeable battery [10].

Figure 1.2: Block diagram of a typical energy harvesting system.

The performance of the entire system depends in particular on the transducer. Its design must be matched to the characteristics of the ambient source as well as to the power need of the target device. A transducer that is well suited to one environment may perform poorly in another, which is why the choice of transduction mechanism is the central design decision in any harvesting application [5].

Designing a good harvesting system is not just about getting the most power out of it. The system must work reliably in real conditions, where the energy source can be irregular or unpredictable. At the same time, it must be small, cheap, and easy to integrate into the target device [5]. Finding the right balance between these requirements is the main challenge in this field, and it has led to the development of many different solutions, as discussed in the following sections.

1.3 Performance metrics

A key challenge in evaluating energy harvesting systems is choosing the right performance metrics. Xu et al.[14] propose two metrics for this purpose. The first is the *power conversion efficiency* (PCE), which measures how much of the power entering the interface circuit actually reaches the output:

$$\mathrm{PCE} = \eta_{\mathrm{Rec}} \cdot \eta_{\mathrm{DC\text{-}DC}} = \frac{P_{\mathrm{out}}}{P_{\mathrm{in}}}. \tag{1.1}$$

The second is the *power extraction efficiency* (PEE), which compares the output power to the maximum power the transducer could deliver under ideal conditions:

$$\mathrm{PEE} = \frac{P_{\mathrm{out}}}{P_{\mathrm{max}}} = \frac{P_{\mathrm{in}}}{P_{\mathrm{max}}} \cdot \mathrm{PCE}. \tag{1.2}$$

The difference between the two is important. A circuit can convert power efficiently (high PCE) while still wasting much of what the transducer could produce, if the circuit itself is not able to work in the optimal operating point. For this reason, PEE is a more useful metric when the goal is to maximise the total power delivered to the load [14].

1.4 Main Transduction Technologies

Mechanical vibrations can be converted into electrical energy through four main transduction mechanisms: piezoelectric, electromagnetic, electrostatic, and triboelectric. Each of them works in a different way and comes with its own advantages and limitations [9, 2].

Piezoelectric

Piezoelectric transducers produce a voltage when a material is mechanically deformed. The most common design is a cantilever beam with a piezoelectric layer on top: when the beam vibrates, the material deforms and generates an electric charge. This approach gives a high output voltage and does not need a magnet or moving parts, which makes it easy to integrate into small devices [1]. The main drawback is that piezoelectric materials are fragile and may crack under large deformations. Also, the output drops quickly when the vibration frequency moves away from the resonant frequency of the beam [10].

Electromagnetic

Electromagnetic transducers generate a voltage by moving a coil near a permanent magnet. This is the same principle used in large electrical generators, applied at a small scale. These harvesters tend to produce a low voltage but can deliver a relatively high current, which makes them a good choice when power output is the main goal. One limitation is that the coil and magnet must move relative to each other, which can be hard to achieve in applications involving large or slowly rotating objects [9].

Electrostatic

Electrostatic transducers work by changing the capacitance between two conductive plates as they move. When the plates get closer or further apart due to vibration, the stored electrical energy changes and can be extracted as useful power. This approach is easy to manufacture using standard microfabrication processes, making it attractive for very small devices. However, it requires an external voltage or a pre-charged material to get started, which adds complexity to the system [9, 2].

Triboelectric

Triboelectric transducers use contact between two different materials to generate surface charges. When the materials touch and then separate, charges build up and can drive a current through an external circuit. This type of harvester, known as a triboelectric nanogenerator (TENG), was first demonstrated by Fan et al.[4] and has attracted a lot of interest because it can be made from cheap, flexible materials and works well at low frequencies, such as those produced by human motion. The main weaknesses are sensitivity to humidity and wear of the contact surfaces over time [8].

Among these four mechanisms, no single technology is universally superior. The best choice depends on the specific application, the characteristics of the available vibration source, and the constraints on size and cost. This comparison motivates the investigation of alternative approaches, including variable reluctance transduction, which is discussed in the following section.

1.5 The Variable Reluctance Principle

Variable reluctance energy harvesting (VREH) is a type of electromagnetic energy harvesting based on Faraday's law of induction. The transducer consists of three main parts: a permanent magnet, a pickup coil wound around a ferromagnetic pole piece, and a toothed ferromagnetic wheel that rotates nearby. When a tooth on the wheel aligns with the pole piece, the magnetic flux through the coil reaches its maximum value. When the gap between two teeth faces the pole piece, the flux drops to its minimum. This periodic variation in flux induces an alternating voltage across the coil, with a frequency that depends on the rotational speed and the number of teeth on the wheel [12]:

$$f_c = \omega\, N_t, \tag{1.3}$$

where ω is the rotational speed in rad/s and N_t is the number of teeth on the weel. Following Faraday's law, the output power of the transducer delivered to a matched load can be expressed as [13]:

$$P_L = \frac{N_{\text{Coil}}^2}{8R_{\text{Coil}}} \Delta_\phi^2 \left[\frac{\omega N_{\text{Tooth}}}{2\pi}\right]^2, \tag{1.4}$$

where N_{Coil} is the number of coil turns, R_{Coil} is the coil resistance, Δ_{ϕ} is the magnetic flux difference between the aligned and unaligned positions, and N_{Tooth} is the number of teeth. This expression shows that the output power grows with the square of the rotational speed, and that it can be increased by maximising the flux difference and the number of coil turns, or by reducing the coil resistance.

What makes VREH different from most other electromagnetic harvesters is that neither the magnet nor the coil needs to move. The flux variation is caused entirely by the relative motion between the toothed wheel and the pickup units. This removes the need for a mechanical connection between the rotating part and the harvester, which is an advantage. It also means that the device is easy to scale: changing the shaft diameter mainly affects the size of the toothed wheel, while the pickup unit can stay the same [13]. In contrast, traditional electromagnetic generators require scaling of the magnets or coils, which becomes costly for large shafts or slow rotational speeds.

At low rotational speeds, the output voltage and frequency are both small, which makes the design of the interface circuit more critical [14]. At the same time, low-speed applications are exactly where VREH has the clearest advantage over other electromagnetic harvesters, since these typically require a relative displacement between coil and magnet that is hard to achieve at low speeds [9].

Its application to energy harvesting is more recent. Kroener et al. [7] were among the first to study a VREH specifically designed for energy harvesting in railway monitoring, showing that the principle could reliably power wireless sensor nodes from the motion of passing train wheels. Since then, the approach has been explored in a growing number of industrial and structural monitoring applications [12, 13].

1.6 State of the Art of VREH

The first applications of the variable reluctance principle to energy harvesting appeared in the context of railway monitoring. Kroener et al. [7] demonstrated that a VREH placed beside a rail track could generate enough power from the passage of train wheels to supply a wireless sensor node, reporting a peak output power of around 5.9 mW at a simulated train speed of 81.5 Km/h. This work showed for the first time that the variable reluctance principle could be practically used as an energy source rather than just a measurement transducer.

A systematic comparison of VREH topologies was later carried out by Xu et al. [12], who surveyed the structures proposed in the literature and evaluated them in terms of output power and power density for low-speed rotating applications. The study identified the *m-shaped* pickup unit, which uses two opposing permanent magnets and a centre pole piece, as the topology with the highest volumetric power density among all structures considered. This result motivated a detailed design study of the m-shaped VREH, in which the influence of three key geometric parameters was investigated: the coil height h_{Coil}, the tooth height h_{Tooth}, and the magnet height h_{M} [13]. The study showed that these parameters involve trade-offs under size constraints, and provided a numerical optimisation method based on finite element analysis (FEA). The optimised structure reached output powers between 0.12 mW and 22 mW at rotational speeds of 5 rpm to 60 rpm within a volume constraint of 43.5 cm ^{3}, which is sufficient to power a range of duty-cycled wireless sensor systems [13].

A practical challenge in VREH systems is that the transducer produces a low-voltage alternating output, which must be rectified and conditioned before it can supply any systems. At low rotational speeds, the output voltage can be well below 1 V, which makes the choice of rectifier circuit critical. Xu et al. [14] compared three passive rectifier topologie: a full-wave bridge rectifier (FWR), a voltage doubler (VD), and a negative voltage converter (NVC), all on the same VREH system at speeds between 5 rpm and 20 rpm. The results showed that the VD achieved the best power extraction efficiency at low speeds, reaching a PEE of 40-60%, while the NVC outperformed it above 18 rpm. The FWR showed the lowest performance across all conditions, with PEE values of only 20-40%. These results confirm that the interface circuit is not just a passive element but actively affects how much power the transducer can deliver to the load.

These studies show that VREH is a mature and well-understood technology for low-speed rotational applications, with demonstrated power levels sufficient for a variety of wireless sensing applications. The most recent step forward in this direction is the multi-phase VREH proposed by Wu et al. [11], which integrates six m-shaped pickup units into a bearing hub unit for large commercial vehicles, reaching output powers between 0.46 W and 4.77 W at rotational speeds of 100 to 400 RPM. This significant increase in output power opens new possibilities for powering more demanding condition monitoring systems. However, it also introduces new challenges in the design of the power conditioning circuit, which must handle six independent AC outputs with different phases and varying voltage levels depending on the rotational speed. These challenges are discussed in the next section.

1.7 Open Challenges

Despite the progress described in the previous sections, several challenges remain open in the design of VREH systems. This section focuses on those that are most relevant to the work presented in this thesis.

Power conditioning for multi-phase outputs

Single-phase VREH systems produce one alternating voltage signal that must be rectified and converted into a stable DC supply. This problem has been studied in detail for low-speed single-phase harvesters, where the choice of rectifier topology has a significant impact on the power extraction efficiency [14].

Many studies have investigated the design of power conditioning circuits for single-phase energy harvesters. The most common approach combines a rectification stage with a DC-DC converter, such as a Buck or Boost circuit, to regulate the output voltage to a level suitable for the target load [15, 16, 6, 3]. However, these solutions are designed for a single AC input and cannot be directly applied to a multi-phase harvester such as the MP-VREH, where six independent AC signals must be processed simultaneously.

Furthermore, many of these solutions do not include a startup circuit or a control system, which are both essential for the reliable operation of the power conditioning circuit in practice. Without a startup circuit, the system is not able to start operating when the harvested voltage is initially too low to power the control electronics. Without a control system, the circuit cannot adapt to changes in the rotational speed or load conditions, which are essential in real-world applications.

This gap in the literature motivates the investigation presented in this thesis.

When the number of pickup units increases, as in the multi-phase VREH proposed by Wu et al. [11], which uses six m-shaped pickup units arranged around a common toothed wheel, the power conditioning problem becomes considerably more complex. Each pickup unit produces an independent AC output with a different phase and the six signals are evenly spaced in time by $60°$ phase shift. This means that standard single-phase rectifier solutions cannot be directly applied, and that the interface circuit must be designed to handle multiple AC inputs efficiently.

Low output voltage at low speed

A fundamental limitation of VREH transducers is that the output voltage scales with rotational speed. At low speeds, the induced voltage may be well below 1 V, which makes rectification difficult because conventional diode bridges lose a significant fraction of the available voltage across their forward drops [14]. In a multi-phase system, this problem affects all six phases simultaneously. Finding a rectifier topology that works efficiently across a wide speed range, from the minimum speed at which the harvester can sustain the load, up to the maximum operating speed, is therefore a key design requirement.

Impedance matching under variable conditions

The maximum power that a VREH can deliver to a load is achieved when the load impedance is matched to the internal impedance of the pickup coil. This matching condition depends on the rotational speed, because the electrical frequency of the output signal changes with speed. For a single-phase harvester, Xu et al. [13] showed that a fixed resistive load leads to an approximately linear increase in output power with speed, whereas a complex conjugate matched load, which compensates for the inductive component of the coil, could further increase the output power, especially at higher speeds. In a six-phase system, achieving dynamic impedance matching for all phases simultaneously adds a further layer of complexity to the interface circuit design.

1.8 Harvester Specifications and Research Objective

The work presented in this thesis is based directly on the MP-VREH proposed by Wu et al. [11]. The electrical characteristics of a single pickup unit, obtained from the transducer model, are summarised in Table 1.1. These values are used throughout this work as the reference operating conditions for the design and evaluation of the proposed power conditioning circuits.

RPM	f (Hz)	V_{Lpk} (V)	R_{coil} (Ω)	L_{coil} (mH)	P_o (W)
100	36.7	0.91	6	21.68	0.41
200	73.3	1.35	6	21.46	0.91
300	110.0	1.59	6	21.27	1.26
400	146.7	1.74	6	21.12	1.51

Table 1.1: Electrical parameters of a single MP-VREH pickup unit at different rotational speeds.

All values in Table 1.1 were obtained under impedance-matched load conditions. V_{Lpk} refers to the peak voltage across the load, and P_o is the average power dissipated in the load.

These challenges motivate the work presented in this thesis, which focuses on the design and evaluation of power conditioning circuits for the six-phase MP-VREH described above. The goal is to identify the interface circuit topology that maximises the power delivered to the load across the relevant operating speed range of 100 to 400 RPM (20-80 Km/h), corresponding to typical driving conditions of large commercial vehicles, for exaple buses.

2 Theoretical Background

This chapter introduces the physical and circuit-level concepts required to motivate the design choices made in the following chapters. In this chapter, starting from Faraday's law, an electrical model of the variable-reluctance energy harvester (VREH) is derived. The model is then used to discuss the maximum power transfer theorem and the role of impedance matching

2.1 Electrical Model of the Variable-Reluctance Harvester

2.1.a Faraday's law

The Faraday's law is the starting point for any electromagnetic transducer, which relates the electromotive force $\varepsilon(t)$ (EMF) induced in a closed loop wire to the time rate of change of the magnetic flux $\Phi(t)$ linked to the loop.

$$\varepsilon(t) = -\frac{\mathrm{d}\Phi_i(t)}{\mathrm{d}t} \tag{2.1}$$

For a coil made by N_t turns sharing the same flux, the total linked flux is $\Lambda(t) = N\,\Phi_i(t)$ and the induced EMF becomes:

$$\varepsilon(t) = -\frac{\mathrm{d}\Lambda(t)}{\mathrm{d}t} = -N_t\frac{\mathrm{d}\Phi_i(t)}{\mathrm{d}t}. \tag{2.2}$$

In case of a coil consisting of N_t turns mechanically spaced one respect to each other leading to different linked flux, the induce voltage can be described by

$$\varepsilon(t) = -\frac{d}{dt}\cdot\sum_{n=i}^{N_t}\Phi_i(t) \tag{2.3}$$

where Φ_i is the flux linkage associated to a single turn of the coil[12].

2.1.b Variable Reluctance Principle

Unlike conventional electromagnetic transducers, in which the EMF is induced by a relative motion between a permanent magnet (PM) and a pickup coil, a variable-reluctance energy harvester (VREH) keeps both the magnet and the coil stationary. The magnetic flux linked to the coil is instead modulated by the passage of a rotating, toothed ferromagnetic wheel in front of a stationary pickup unit: as a tooth or a slot alternately faces the pickup unit, the reluctance of the magnetic circuit changes periodically, and it is this reluctance variation – rather than any bulk motion of the magnet – that drives the flux change $\Phi_i(t)$ in (2.3) [12]. Because the rotating part carries no magnet, no coil and no electrical connection, the resulting transducer is mechanically simple, robust, and easily rescaled to shafts of different diameters [13].

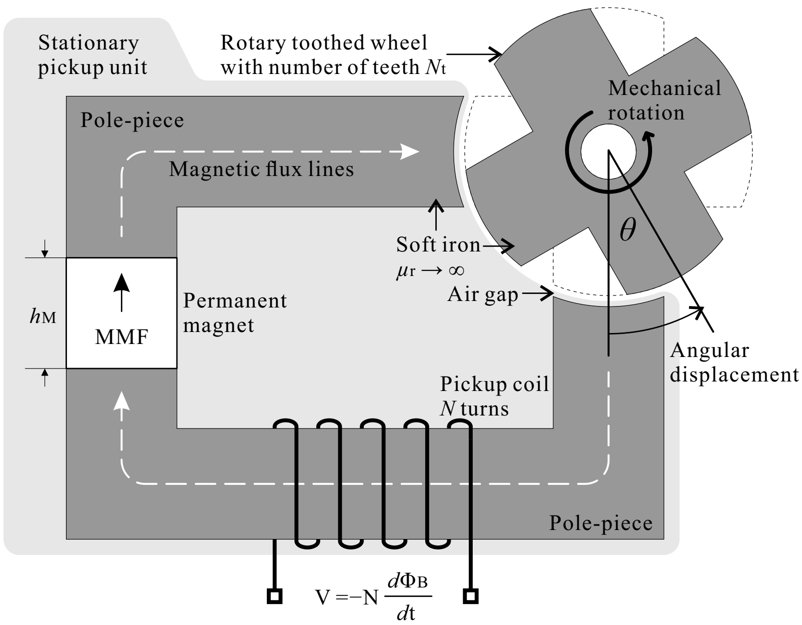

Figure 2.1: Structure of a variable-reluctance sensor: a stationary pickup unit, composed of a permanent magnet, a soft-ferromagnetic pole-piece and a pickup coil, faces a rotating toothed wheel across an air gap.

The pole-piece and the toothed wheel are both made of soft ferromagnetic material, while the PM is a hard ferromagnetic material providing a magneto-motive force (MMF) $\mathcal{F}_M$ that can be taken as constant. As the wheel rotates, the pickup unit cycles between two extreme relative positions: the *aligned* position, where a pole of the pickup unit faces a tooth of the wheel and the air-gap reluctance is minimum, and the *unaligned* position, where the pole faces a slot and the reluctance is maximum [12, 13]. The corresponding flux swings between a maximum Φ_{max} and a minimum Φ_{min}, so that the relevant design quantity is the flux difference $\Delta\Phi = \Phi_{max} - \Phi_{min}$ between the two extremes.

From flux swing to output power

For a wheel with N_t teeth rotating at angular velocity ω, the electrical frequency of the induced voltage is

$$f_c \;=\; \omega\, N_t. \tag{2.4}$$

At low rotational speeds the coil impedance is dominated by its resistance R_{Coil}, so that maximum power is transferred to a purely resistive load matched to it, $R_L = R_{Coil}$; under this condition the load voltage is related to the open-circuit coil voltage $V_c(t)$ of (2.3) by

$$V_L(t) \;=\; \frac{1}{2}\, V_c(t). \tag{2.5}$$

Assuming the time-varying flux to be an ideal sinusoidal wave, uniformly distributed in the N_c turns of the coil, the peak-to-peak voltage developed across the matched load follows from (2.3) and (2.5) as [12]

$$V_{L,pp} \;=\; -\frac{1}{2}\, N_c\, \frac{\Phi_{max} - \Phi_{min}}{\frac{1}{2}T_c} \;=\; -N_c\, f_c\, \Delta\Phi, \tag{2.6}$$

where $\frac{1}{2}T_c$ is the time taken by the wheel to rotate from the aligned to the unaligned position, i.e. from a tooth to a slot centred on the pole-piece.

Because $V_{L,pp}$ in (2.6) is the peak-to-peak amplitude of an (assumed) sinusoidal load voltage, its root-mean-square value is $V_{L,rms} = V_{L,pp}/(2\sqrt{2})$, so that the RMS power dissipated in the matched load is [12]

$$P_L \;=\; \frac{V_{L,rms}^2}{R_L} \;=\; \frac{V_{L,pp}^2}{8\, R_{Coil}} \;=\; \frac{(\omega\, N_t\, N_c)^2}{8\, R_{Coil}}\, (\Delta\Phi)^2, \tag{2.7}$$

where the last equality follows from substituting (2.4) into (2.6). Equation (2.7) shows that the harvested power scales quadratically with the rotational speed ω, the flux swing $\Delta\Phi$, and the number of coil turns N_c, and linearly with the number of teeth N_t, while it is inversely proportional to the coil resistance R_{Coil}.

For a given application (i.e. fixed ω and N_t), the design of the pickup unit

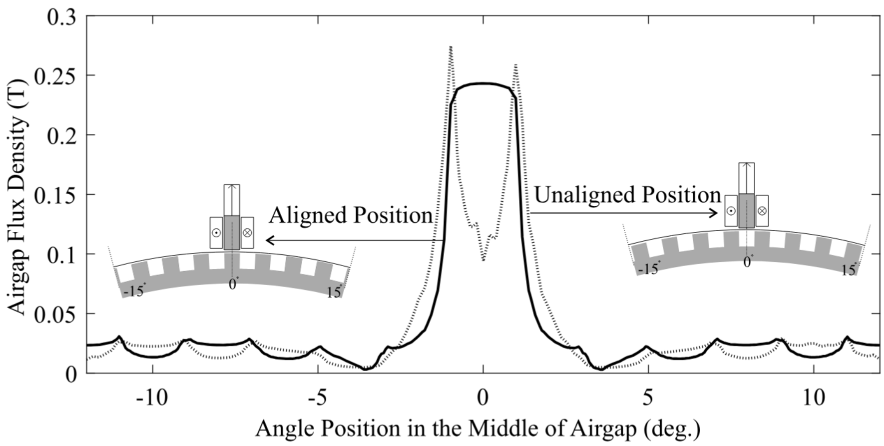

Figure 2.2: Airgap flux density distribution for aligned and unaligned positions. Reproduced from [12].

therefore splits into two largely independent optimization problems [13]:

$$P_L \propto \frac{N_c^2}{R_{Coil}}, \tag{2.8a}$$

$$P_L \propto (\Delta\Phi)^2 = \left(\frac{\mathcal{F}_M}{\Delta\mathcal{R}}\right)^2, \tag{2.8b}$$

where $\Delta\mathcal{R} = \mathcal{R}_u - \mathcal{R}_a$ is the difference between the total loop reluctance at the unaligned and aligned positions. Sub-function (2.8a) is a winding-design problem: for a coil of resistivity ρ_{wire} occupying a cross-section of height h_{Coil} and width W_{Coil} inside the pole-piece, it can be shown that (2.8a) reduces to [13]

$$P_L \propto \frac{h_{Coil}\, W_{Coil}\, f_w}{\rho_{wire}}, \tag{2.9}$$

with $f_w < 1$ the winding fill factor of the coil; for a fixed available cross-section, output power is thus maximized simply by maximizing the copper fill of the winding window. Sub-function (2.8b), in contrast, is a magnetic-circuit design problem: with the MMF $\mathcal{F}_M = H_c\, h_M$ fixed by the magnet coercivity H_c and its height h_M, maximizing $\Delta\Phi$ reduces to maximizing the reluctance swing $\Delta\mathcal{R}$ between the two extreme rotor positions.

Magnetic-circuit model of the reluctance swing

The reluctance of a homogeneous flux-path element of length l, cross-section A and relative permeability μ_r is

$$\mathcal{R} \;=\; \frac{l}{\mu_0\,\mu_r\,A}. \tag{2.10}$$

Since the soft ferromagnetic pole-piece and toothed wheel have a relative permeability several orders of magnitude larger than that of air ($\mu_r \approx 1$) or of the PM ($\mu_r \approx 1.05$), their contribution to the loop reluctance is negligible, and $\Delta\mathcal{R}$ is dominated by the air gap and by the magnet's own reluctance $\mathcal{R}_M = h_M/(\mu_0\mu_M A_M)$ [13]. Denoting by $\mathcal{R}_{g,a}$ and $\mathcal{R}_{g,u}$ the air-gap reluctance at the aligned and unaligned positions, respectively, the flux at each extreme follows from Hopkinson's law as $\Phi_{max} = \mathcal{F}_M/(\mathcal{R}_{g,a}+\mathcal{R}_M)$ and $\Phi_{min} = \mathcal{F}_M/(\mathcal{R}_{g,u}+\mathcal{R}_M)$, so that the flux swing of (2.8b) can be written explicitly as

$$\Delta\Phi \;=\; \mathcal{F}_M\left(\frac{1}{\mathcal{R}_{g,a}+\mathcal{R}_M} - \frac{1}{\mathcal{R}_{g,u}+\mathcal{R}_M}\right). \tag{2.11}$$

Equation (2.11) makes explicit that, once the magnet material and the air-gap length are fixed by mechanical constraints, $\Delta\Phi$ is governed by the pickup-unit geometry surrounding the air gap, namely the tooth height h_{Tooth} (which sets $\mathcal{R}_{g,u}$, since a taller tooth increases the reluctance seen at the unaligned position up to a saturating knee at approximately $h_{Tooth} \approx 3$ mm) and the magnet height h_M (which enters both $\mathcal{F}_M$ and $\mathcal{R}_M$, so that an optimal, non-trivial value of h_M exists) [13].

For a fixed available volume, expressed as a hollow cylinder of overall diameter d_o, shaft diameter d_{Shaft} and depth l_o, the volumetric power density of the harvester is

$$p_v \;=\; \frac{P_L}{V_{VREH}}, \qquad V_{VREH} \;=\; \frac{\pi}{4}\,l_o\,(d_o - d_{Shaft})^2, \tag{2.12}$$

and, once the pole pitch and slot width are fixed, the available radial space $C = h_{Coil} + h_{Tooth}$ must be split between the coil and the tooth. Since P_L increases monotonically with h_{Coil} through (2.9) but only up to the knee point through h_{Tooth} in (2.11), for a given C there is an optimal h_{Coil}/h_{Tooth} ratio that maximizes p_v [13]. Similarly, setting $\frac{\mathrm{d}\Delta\Phi}{\mathrm{d}h_M} = 0$ in (2.11) yields a closed-form optimal magnet height,

$$h_M^* \;=\; \mu_M\,\frac{k_2}{k_1}\,\frac{W_{g,a}}{W_{g,u}}\,\sqrt{l_{g,a}\,l_{g,u}}, \tag{2.13}$$

where $l_{g,a}$, $l_{g,u}$, $W_{g,a}$, $W_{g,u}$ are the effective air-gap length and width at the aligned and unaligned positions, and $k_1, k_2 > 1$ are correction factors accounting for flux leakage and for the MMF drop in the finite-permeability soft iron, respectively [13].

Pickup-unit topology and flux-improvement strategies

A comparative survey of pickup-unit structures used in VR sensors identifies three complementary strategies to increase $\Delta\Phi$ for a given MMF [12]:

1. providing a soft-ferromagnetic return path for the flux, reducing leakage outside the useful magnetic circuit;
2. placing both the PM and the pickup coil as close as possible to the air gap, improving their coupling to the rotating wheel;
3. using multiple PMs in a repelling arrangement, so that their flux loops are forced to converge through a shared pole – since $\Delta\Phi$ enters (2.7) quadratically, combining flux loops in this way is more effective than simply adding them.

These strategies were compared quantitatively by finite-element analysis (FEA) on seven pickup-unit structures, all evaluated on the same 270 mm-diameter shaft rotating at $\omega \approx 1.06$ rad/s (≈ 10.1 rpm) [12]. The most basic, single-magnet structure without a return path (taken as the reference) produced only 15.74 μW; adding a ferromagnetic return path and moving the PM adjacent to the air gap already yields up to a 17.56-fold improvement, while combining all three strategies in an m-shaped pickup unit with two repelling PMs – with the pickup coil wound around the shared centre pole – reaches over 840 μW, i.e. a 53-fold increase in output power and a 30-fold increase in power density (about 158 nW/mm^3, the highest of all structures compared) over the reference, and roughly 20% more than the next-best design [12]. This gain, however, comes at the cost of the largest PM volume among the compared structures and therefore the highest parasitic (cogging) torque ripple, whereas structures with smaller, single or non-repelling magnets remain more torque-friendly at the expense of lower power density [12].

In this m-shaped topology, the two PMs face the air gap with the same magnetic pole, so that their fields are mutually repelling. At the aligned position, each outer pole is coupled to a tooth of the wheel through a low-reluctance path, and the two flux loops originating from the two magnets are forced to converge through the centre pole, producing a high flux density in the region spanned by the coil. At the unaligned position, both outer poles face a slot, the air-gap reluctance reaches its maximum, and the flux collapses accordingly [13].

A detailed structural optimization of this topology, validated experimentally with FEA-to-measurement errors below 3% over several magnet-, tooth- and coil-height combinations, confirmed that a size-constrained m-shaped VREH (volume $\approx 43.5\,\mathrm{cm}^3$) can deliver between $0.12\,\mathrm{mW}$ and $22\,\mathrm{mW}$ over the 5–60 rpm range typical of low-speed industrial shafts [13]. This m-shaped, dual-magnet pickup unit is the reference topology adopted for the harvester considered in this thesis.

It is worth noting that (2.7) neglects the coil inductance, an approximation only valid at sufficiently low electrical frequencies. In the optimized structure discussed above, the coil presents a non-negligible inductance ($L_{Coil} \approx 210\,\mathrm{mH}$); as ω increases, the corresponding inductive reactance becomes comparable to R_{Coil}, and a purely resistive load match, $R_L = R_{Coil}$, no longer maximizes the extracted power. Experimentally, the effect of a complex-conjugate impedance match (a series-compensating capacitor added to the resistive load) was found to be negligible below $\omega \approx 2.1$ rad/s ($\approx 20\,\mathrm{rpm}$), but above this speed a purely resistive load increasingly fails to exploit the harvester's capability, so that a full complex-conjugate match between coil and load is required to reach the maximum output power of 22 mW quoted above at the top of the evaluated speed range [13]. This observation motivates the discussion of impedance matching under variable operating conditions developed later in this chapter.

3 Rectifiers

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Wish to say “Thank you”? Write it in *docs/acknowledgements.txt*.

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